



Webfleet TPMS

24/7 tyre control: React fast. Eliminate downtime.

24/7 tyre control: Reliable road safety.



TPMS: Tyre Pressure Monitoring System

Webfleet TPMS checks your tyre pressure and temperature in real time. With predictive tyre management, problems are detected before they lead to costly repairs or downtime. This helps you to cut running costs, improve road safety and reduce your impact on the environment. And, with less chance of breakdowns, there's less chance of disruption.

Whatever you're transporting—passengers or cargo—TPMS helps you ensure it gets to its destination safely and on time.

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The background

- What is TPMS?
 - TPMS stands for Tyre Pressure Monitoring System. It's a system fitted on a vehicle that constantly evaluates the pressure of the tyres. It then transmits corresponding information and alerts in real time to the driver and/or fleet manager, even while the vehicle is running.¹
- Legislation
 - EU regulation [EC 661/2009](#) mandates that TPMS must be installed to all passenger cars from November 2014.
 - From July 2022, [EU 2019/2144](#) stipulates all new commercial vehicle types approved (light commercial vehicle and heavy commercial vehicles) will be required to have TPMS as standard to pass the homologation process².
 - From July 2024, as part of [EU 2019/2144](#) all commercial vehicles registered (not just new models, but also existing models that are still in production) will be required to have TPMS as standard.
 - EU directive [2014/45/EU](#) lists TPMS as part of the criteria for periodic roadworthiness testing: systems must be operational.
 - NOTE: EU directives are often implemented differently by member states—it is currently unknown if all member states will test TPMS systems.

¹ Source: [EC](#).

² The type of TPMS that must be added by vehicle manufacturers is similar to that already implemented in all passenger cars in Europe today. It could be a simple warning light in the dashboard when pressure loss is detected. "Vehicles shall be equipped with an accurate tyre pressure monitoring system capable, over a wide range of road and environmental conditions, of giving an in-vehicle warning to the driver when a loss of pressure occurs in a tyre."

Opportunity

- The regulation works in our favour, as it makes TPMS in Heavy Commercial Vehicles (HCVs) an even more relevant topic. Our solution anticipates the application of UN ECE R141.
- Combining market-leading products from Webfleet and Bridgestone to create digital solutions focused on tyres benefits customers while adding value to existing propositions such as Fleet Care tyre contracts.
- Webfleet's data-oriented approach goes beyond notifications and alerts, providing real-time insights that enhance customers' efficiency in fleet management.
- While the changes in legislation relate to new vehicles only, there is a huge legacy vehicle parc. In Europe, there are around 8,195,907² HCVs in use (trucks and buses). Most of these vehicles do not have TPMS and represent a big part of our target market.
- Our TPMS solution is designed to be data-source agnostic.
 - We offer an aftermarket solution with our own dedicated TPMS hardware.
 - We offer a solution for new vehicles that already have TPMS fitted by the OEM (and complying with UN ECE R141), using our Direct FMS³ service.
 - We offer solutions based on OEM.connect where manufacturers support TPMS data requirements.

¹ Source: Frost and Sullivan (2018)

² ACEA's 'Vehicles on European Roads' report - FEBRUARY 2024

³ Direct FMS is an additional service. The subscription fee is not included in the existing Webfleet tariff. Direct FMS enables additional features in Webfleet when FMS is not available in the vehicle.

Different applications, same intuitive interface

To address a fleet's unique needs, we offer advanced solutions tailored for multiple vehicle types and applications. Our comprehensive range includes:

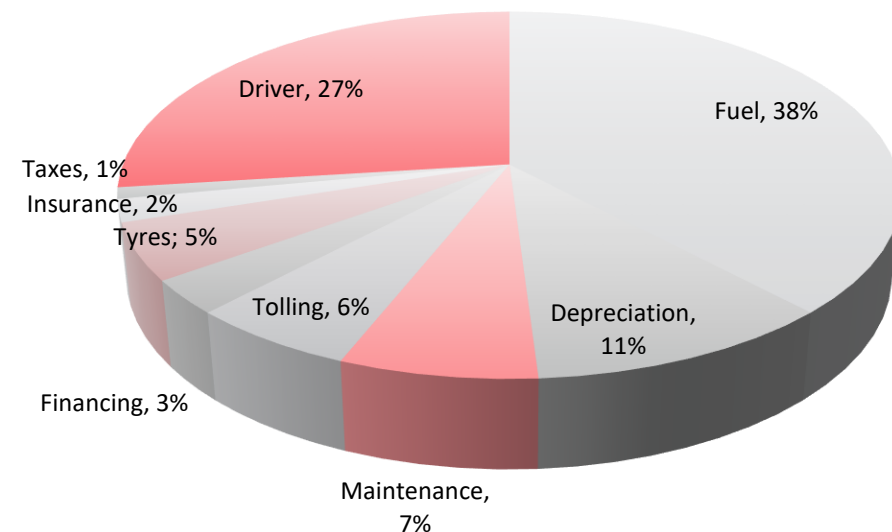
1. **Manufacturer (OE) TPMS:** uses OEM receivers and sensors, connectivity to LINK 7x0 and FMS
2. **Trailer-only TPMS:**
 - » **Aftermarket TPMS for trailer only:** LINK 350 and strap-mounted sensors
 - » **Manufacturer (OE) TPMS for trailer only:** LINK 350 and OEM sensors
 - » **OEM.connect for trailer only:** no hardware required (cloud-to-cloud integration with OEM.platform)
3. **Aftermarket TPMS:** uses LRX 100 and valve-mounted sensors

While the hardware and data sources vary across TPMS solutions, the clean WFS UX/UI remains consistent.

Functionalities not applicable or only applicable to specific TPMS methods are highlighted in the following chapters.

Why maintaining correct tyre pressure is important

- Tyres make up 5% of the average fleet's TCO (total cost of ownership).*
- Poor tyre pressure management can negatively impact finances and safety in numerous ways, including:
 - Miles/kilometres not delivered by the tyre
 - Higher tyre disposal and/or recycling costs
 - Missed opportunity to reuse/retread the tyre
 - Increased fuel consumption
 - Increased vehicle breakdown (downtime) and incidents costs
 - Decreased driving safety
- To get the most out of fuel-efficient tyres on the road, it's important to maintain the correct tyre pressure.

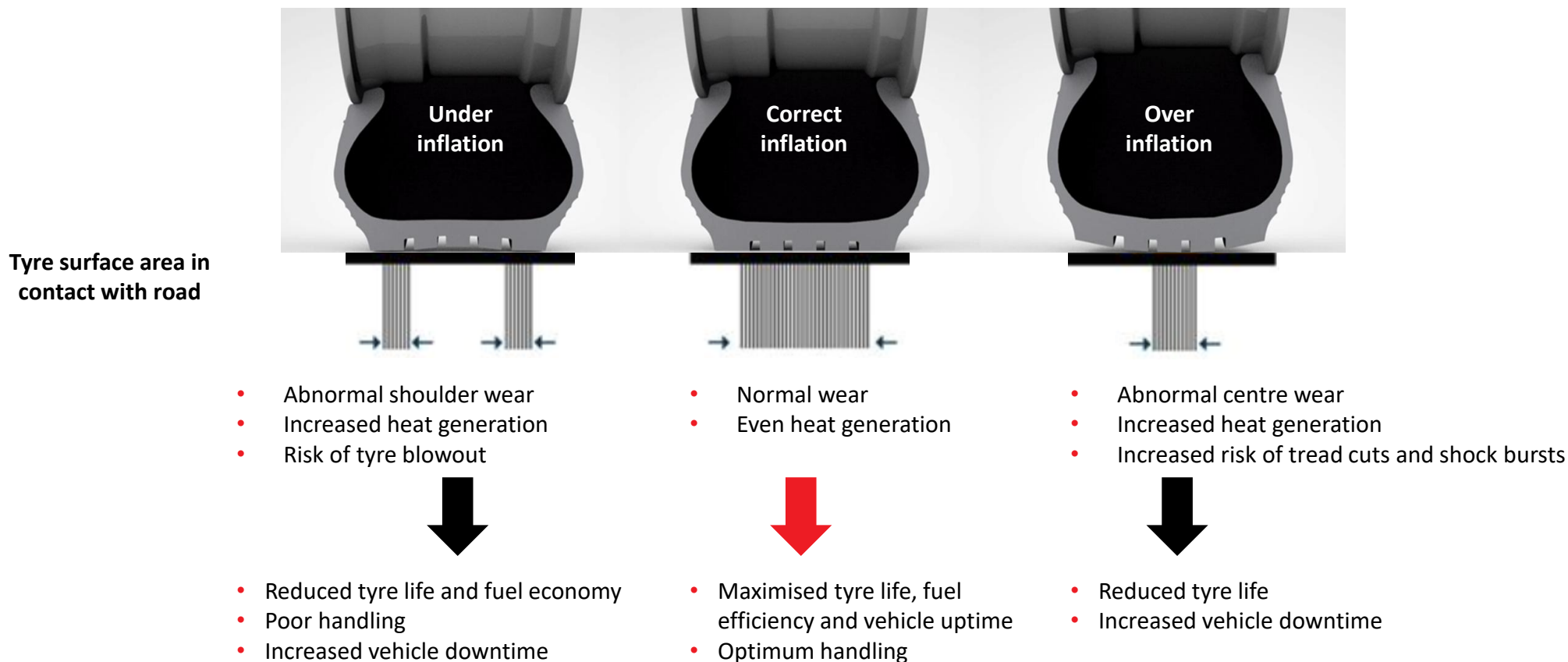


**Annual TCO (total cost of ownership)
for HCV > 12t in Europe***

* Source: Connected Fleet Services Global Study 2018, PTOLEMUS Consulting Group. Hypothesis: Average mileage of 120,000km per year, tyres: 14 tyres, EU400 per tyre, lifetime of 90,000km.

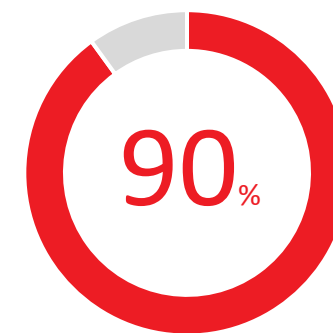
Why maintaining correct tyre pressure is important

- A tyre is a flexible structure filled with compressed air. Inflation pressure plays an important role in carrying the vehicle load. It also impacts driving safety, vehicle downtime and operational costs.



Why maintaining correct tyre pressure is important

- Tyres are the main causes of vehicle breakdowns
 - 87% of heavy truck breakdowns on English trunk roads and motorways are tyre-related.¹
 - This leads to an average two hours of downtime per tyre-related incident.
- Under-inflated tyres increase rolling resistance², resulting in higher fuel consumption
 - A tyre inflated to 9 bar falls back to 7 bar (20% under-inflation) after 16 weeks.
 - This translates to a 2.5% increase in fuel consumption.³
- Poorly inflated tyres negatively affect braking distance and vehicle handling control
 - In England, 52.5% of fatal or seriously-injured accidents associated with vehicle defects were attributed to 'Tyres illegal, defective or under inflated'.⁴
- Poor tyre pressure management reduces tyre service life (tyre mileage)
 - A 20% under-inflated tyre will only last 75% of its regular service life.⁵



90%
of tyre related breakdowns
are caused by slow leaks

**And yet, 20% of commercial
tyres on the road are
significantly underinflated⁵.**

¹ Source: [RHA](#), UK

² The rolling resistance of a tyre is the sum of forces (e.g., friction, air resistance) required purely to roll it.

³ Internal data from Bridgestone. The impact may be more significant depending on the tyre specifications. 20% under-inflation can increase fuel consumption up to 5% in some older generation tyres.

⁴ Source: [Highways England SRN Casualty Report 2018](#)

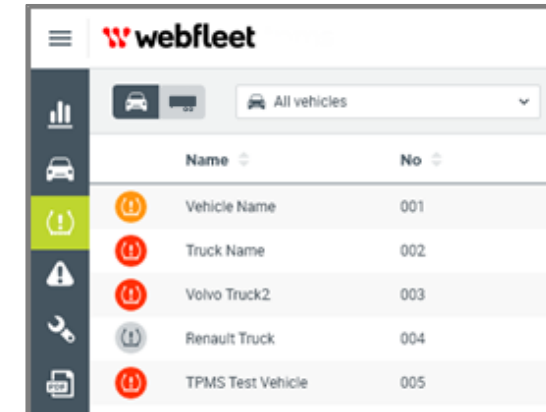
⁵ Source: Bridgestone

Highlights

- Dedicated Webfleet TPMS view
 - Clear overview of all TPMS-equipped vehicles and trailers with the latest tyre pressure status
 - Notifications for key TPMS-related events, e.g. pressure loss or high temperature
 - Automatically created maintenance tasks for less time-critical events
- Smart TPMS sensors
 - A valve-based sensor that is small, lightweight and ideal for the aftermarket. It can be attached to a standard valve on a heavy commercial vehicle without the need to remove the tyre.
 - Various adapters to enable fitting to most wheel configurations
- Compatible with vehicles fitted with TPMS from the manufacture
 - Vehicles complying with UN ECE R141 do not need additional sensors, as the sensors fitted from the OEM are used for Webfleet's TPMS solution.
- Webfleet TPMS Tools app support
 - Designed to be used in the workshop to assist in maintaining the correct operation of Webfleet TPMS
- Driver warning options
 - Simple notifications of key TPMS events for all current PRO Driver Terminals¹
 - An in-cab indicator to alert the driver that tyre pressure is low
- Vehicle support
 - TPMS is available for bus/coach, rigid bodied trucks, trailers and tractor/trailer combinations.

¹ Available for PRO 2020, PRO 7350, PRO 8375, PRO 8475 and Work App

TPMS view in Webfleet



Name	No
Vehicle Name	001
Truck Name	002
Volvo Truck2	003
Renault Truck	004
TPMS Test Vehicle	005

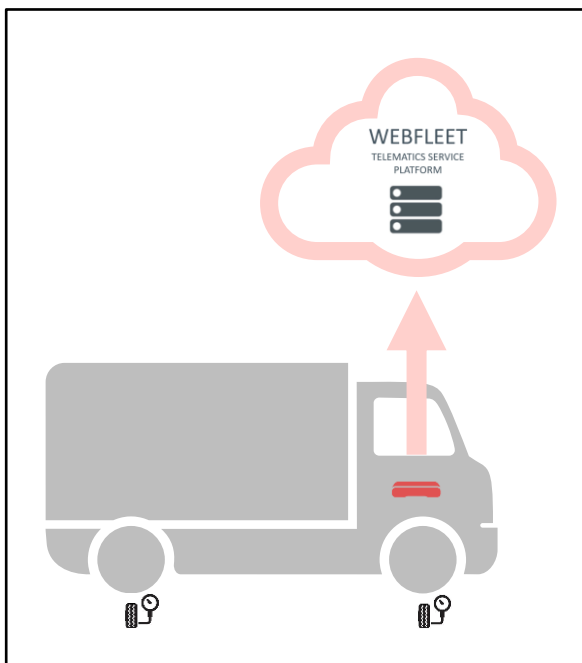
Webfleet TPMS Tools application



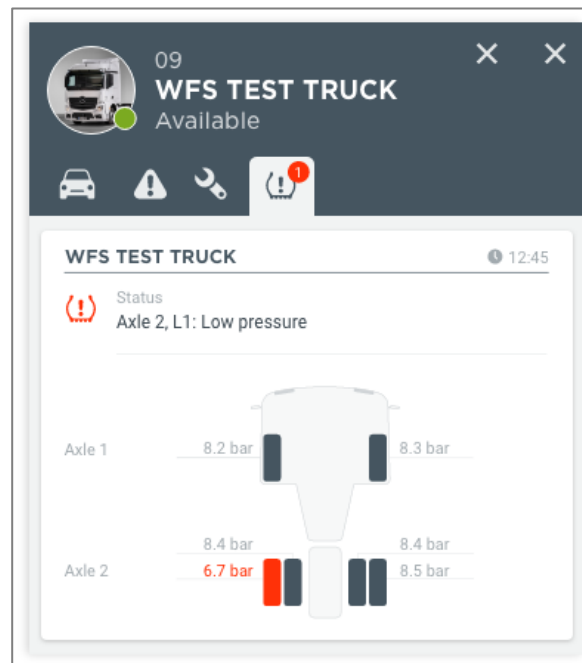
TPMS notifications in Driver Terminal/Work App



Webfleet TPMS: How it works



- 1 Webfleet TPMS reads sensor data* from each tyre. Tyre pressure levels and temperature readings are sent to the Webfleet cloud.



- 2 The system analyses sensor data to identify tyre pressure and temperature issues via notifications in Webfleet.



- 3 Both fleet manager and driver (optional) receive a real-time alert to take action before costly repairs or downtime are needed.

* Sensor data may come from Webfleet's own aftermarket sensors or sensors fitted by the vehicle manufacturer.

Challenges

- Tyres are a vehicle's first and only contact with the road. Poor tyre management can lead to serious challenges.



Downtime

Tyre-related incidents cause significant disruption to business efficiency and operational output, increasing unexpected downtime and costs.



Service interruptions

Tyre-related incidents can impact customer service level agreements and damage brand reputation. Penalty fines, spoilt stock and lost opportunity costs can all quickly escalate.



Driver safety and retention

Poor tyre pressure affects braking distance and vehicle handling, putting drivers and other road users at risk. Stress and injury caused by poor driver safety can negatively impact driver recruitment and retention.



Fuel consumption

Increased rolling resistance due to poor tyre pressure or slow pressure leaks leads to higher fuel consumption and increased CO2 emissions.



Maintenance and repairs

For many businesses, potentially avoidable tyre-related incidents lead to higher repair costs and higher service charges.



Tyre performance

Poor tyre pressure reduces mileage performance, increases tyre wear and leads to a reduction of tyre life. This reduces performance and return on investment.

Value proposition at a glance

Target audience	Pain points	Benefits	Proof points and value messages
Fleet type: Trucks <i>(Tractor/rigid)</i>   Buses and coaches  Trailers (standalone)  Tractor/trailers  Decision maker: Maintenance Manager Safety Manager Fleet Manager CFO/CEO	- How do I prevent tyre incidents happening? - When tyre incidents do occur, I want my drivers to get to a service point quickly. - I spend too much time on manual tyre maintenance inspections.	Maximise vehicle uptime	Fewer interruptions, less downtime - Spots issues that are difficult to detect without TPMS, such as slow pressure leaks - Reduces the chance of tyre failure that can disrupt your service and lead to delays - Allows you to plan more flexibly with timely information Better maintenance - Maintains tyres more efficiently with real-time pressure monitoring - Reduces time spent on manual tyre inspections - Lowers the risk of tyre failure with targeted tyre maintenance
	I want my vehicles and drivers to be safer on the road.	Enhance road safety	Less risk of accidents Tyres are easily kept at the correct pressure so tyre performance remains high, keeping vehicle braking and handling control consistent.
	- Tyre incidents cost me too much time and money. - I want to bring down my overall running costs.	Lower running costs	Lower fuel and breakdown costs - Properly inflated tyres save you money. For example, a tyre under-inflated by 20% will use 2.5% more fuel than a properly inflated tyre. - Reduces tyre breakdowns by detecting tyre issues beforehand Reduces tyre cost per kilometre - Correct tyre pressure helps maximise tyre mileage by improving retreadability. Retreading a tyre gives it a 'second lifetime', and is less expensive than a new tyre. Optimises tyre life - Correct tyre pressure avoids excessive wear and tear.
	I want to run a more sustainable business.	Increase sustainability	Reduces CO2 emissions - By improving your fuel economy, TPMS helps lower your emissions. Reduces tyre waste - Correct tyre pressure helps optimise the lifetime of your tyres, consuming less resources.

Overview



Live view of your tyres

See all your TPMS-equipped vehicles and trailers with latest tyre status. You can access the details of an individual tyre (sensor).



Slow leak identification

Spots slow pressure leaks, before a driver possibly could. This reduces the risk of developing dangerously low-pressure levels.



In-cab driver alerts

Get a simple notification of critical tyre issues displayed on a Driver Terminal², via an in-cab indicator³ and/or Work App.



Tyre issues display

Each vehicle or trailer tyre with an issue is colour-coded to indicate the severity of the issue: amber for warnings, red for critical alarms.



Tracking and mileage display

When TPMS is used in combination with a vehicle tracking device¹, you can see the vehicle's locations and odometer.



TPMS Tools app

Our dedicated application is designed for workshop use to ensure the sensors are shown in the correct wheel position in Webfleet.



Real-time notifications

Receive a notification immediately when an abnormal tyre pressure or temperature issue is detected.



Trailer support

Available with full track and trace functionality, or simplified⁴ TPMS only functionality where both truck and trailer are fitted with compatible TPMS equipment.

¹ Compatible with a LINK 710 or LINK 740

² Available for PRO 2020, PRO 7350, PRO 8375 and PRO 8475

³ Availability to be announced

⁴ TPMS notifications only sent when connected to the towing truck

Dedicated TPMS module

In Webfleet

1. A new main menu view displays TPMS functions, including:

- Vehicles equipped with TPMS
- Trailers equipped with TPMS
- Licence plate
- Coupled trailers/vehicles
- Missing sensors
- Number of warning issues
- Number of alarm issues
- Last issue

2. Select from TPMS equipped vehicles or trailers

The screenshot displays the Webfleet TPMS interface. The main menu on the left (labeled 1) includes icons for vehicles, trailers, and TPMS. The main table (labeled 2) lists vehicles and trailers with columns for Name, No., Number plate, Assets, Missing sensors, Alarms, and Warnings. The 'WFS TEST TRUCK' is highlighted. On the right, a detailed view of the truck shows tire pressure gauges and status indicators.

Name	No.	Number plate	Assets	Missing sensors	Alarms	Warnings
Alex Foster	01	GG337OL	—	6	—	12
Ania Kowalski	156	UZZ51AA	131 - TRAILER 01	—	—	—
Arturo Sanchez	158	BL721AZ	132 - TRAILER 02	4	—	4
Anneke de Vries	159	OAS12SD	—	—	—	2
Catarina Costa	016	XK983HD	—	—	—	1
Carlos Torres	019	HX123LL	—	—	—	—
Chloe Martin	021	LFR123X	—	—	—	—
Christoph Maier	02	WFS34X	—	—	—	—
Ethan Venter	03	HJU345FG	—	—	—	—
Francisco López	24	XCD567HJ	—	—	—	—
John Garcia	10	HGV764KJ	—	—	—	—
Liam Smith	25	FST354WE	—	1	—	1
WFS TEST TRUCK	09	JHF653ER	141 - TRAILER 03	1	—	1
Oliver Brown	06	HK987FO	—	—	—	2
William Davis	181	LOJ456UY	086 - TRAILER 04	—	2	—

The right sidebar shows a detailed view of the 'WFS TEST TRUCK' with a status indicator and a diagram of the truck's axles and tires, displaying pressure readings for each axle.

Dedicated TPMS module

In Webfleet

3. Trailers equipped with TPMS are displayed in a separate “Trailers” tab showing:

- Trailers equipped with TPMS
- Licence plate
- Coupled vehicles
- Missing sensors
- Number of warning issues
- Number of alarm issues
- Last issue

The screenshot displays the Webfleet TPMS interface. A red circle highlights the 'Trailers' tab, which is the third icon in the left sidebar. The main table lists assets with columns for Name, No., Number plate, Vehicle, Missing sensors, Alarms, and Warnings. Asset ASSET05 is highlighted in green. To the right, a detailed view for ASSET05 shows its status as 'Standstill' and a pressure gauge for Axle 1 and Axle 2, both at 9.0 bar.

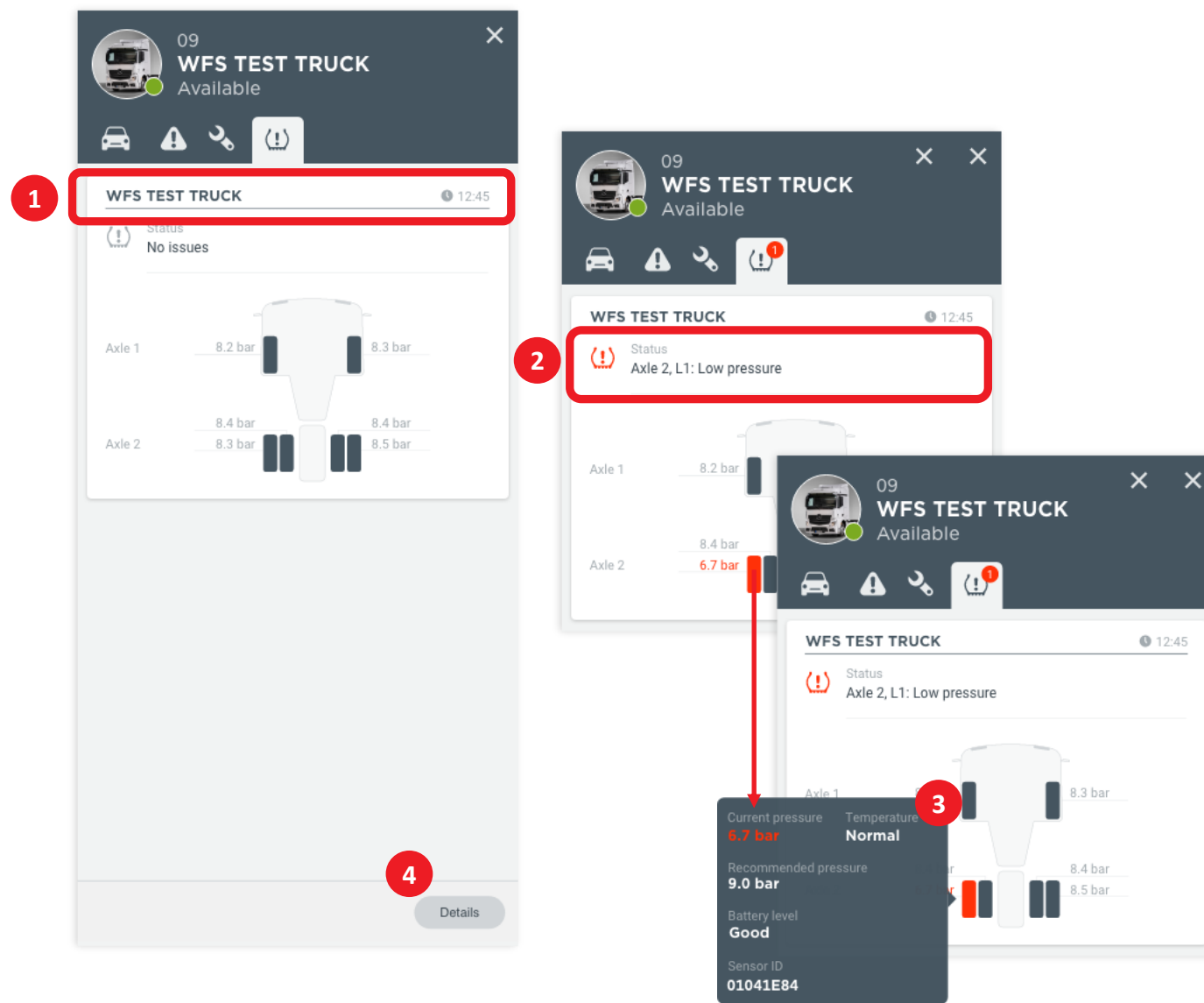
Name	No.	Number plate	Vehicle	Missing sensors	Alarms	Warnings
ASSET01	093	ASSET012	—	—	—	—
ASSET04	098	ASSET04	—	—	—	—
ASSET05	102	ASSET05	—	—	—	—
ASSET06 - Test_0004	106	ASSET06	+ 1	—	—	—
ASSET07 - Test_0004	107	ASSET07	+ 1	—	2	1
SDEV18_Mass_Sim_Asset#1	SDEV18#2	WWL SDEV18 #2	+ 1	—	—	1
SDEV18_Mass_Sim_Asset#2	SDEV18#3	WWL SDEV18 #3	+ 1	2	—	2

ASSET05 Details:
 Status: No issues
 Axle 1: 9.0 bar
 Axle 2: 9.0 bar

Detailed vehicle/trailer view

In Webfleet

1. TPMS data is sent with the vehicle update. The time of the most recent TPMS data is displayed on the interface.
2. The current tyre issue is displayed with a clear text description. If no issue is detected, this is also indicated.
3. The last known tyre pressures are always displayed on the vehicle chassis. You can see clearly whether your tyres are correctly inflated.
 - Tyre sensors detect a wide range of pressure and temperature related events. Each tyre with an issue is colour-coded to indicate the severity of the issue.
 - When you hover the mouse over the problematic tyre, a tool tip window displays further information about the tyre (sensor).
4. The “Details” button allows you to view more information about the sensors.



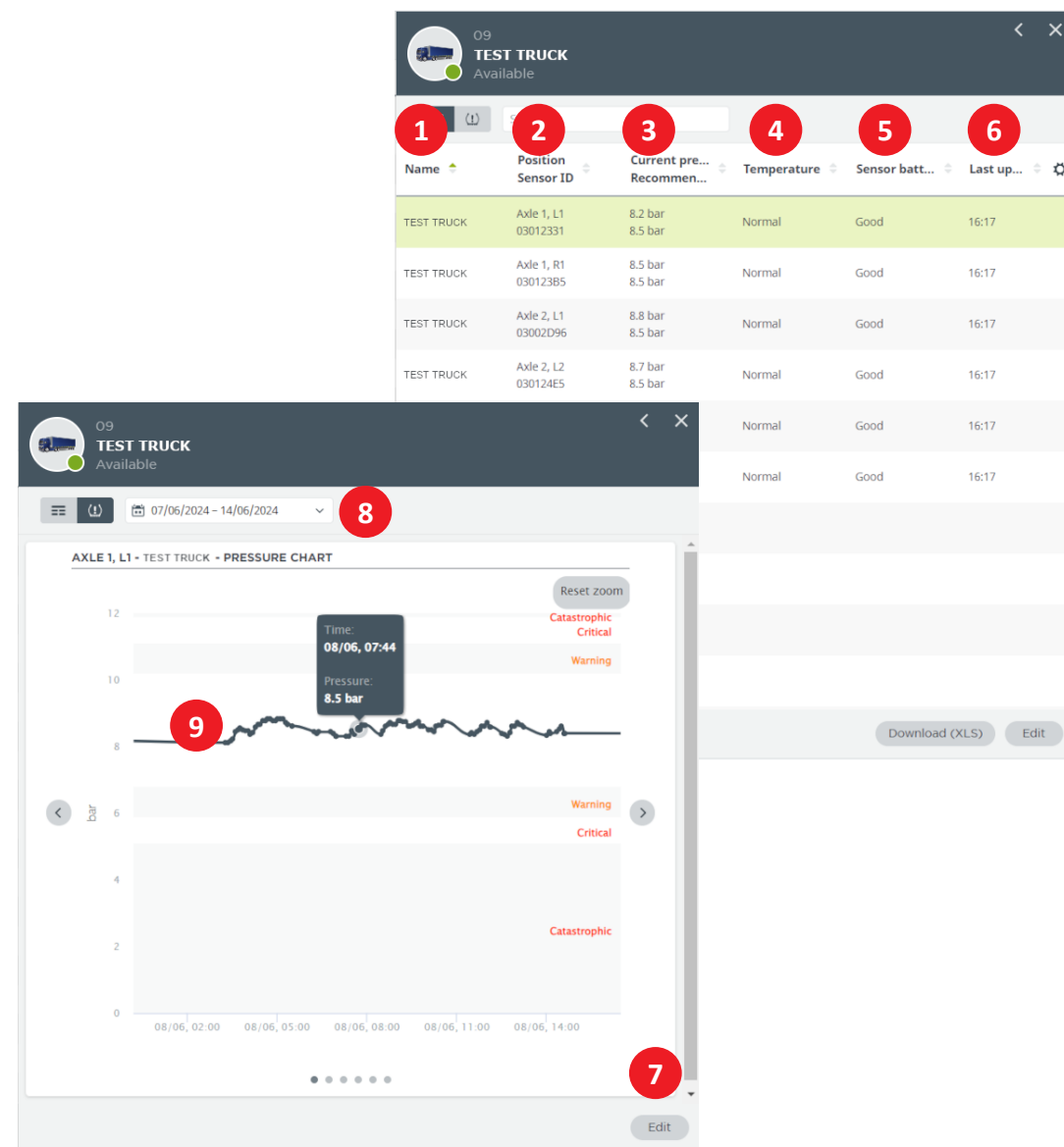
Sensor details view and pressure charts

Under Vehicle view

The sensor detail's view allows you to easily see the pressure history of your tyres as well as all current information about each sensor fitted to a specific vehicle, including:

1. Vehicle name (truck or trailer)
2. Sensor position and identification number⁽¹⁾
3. The current and recommended tyre pressures
4. The temperature status of the tyre (sensor)
5. The current battery level of the sensor
6. Time the TPMS data was last updated
7. Click here to [edit the sensors](#)
8. Date selector for pressure chart
9. Tyre pressure over timeline

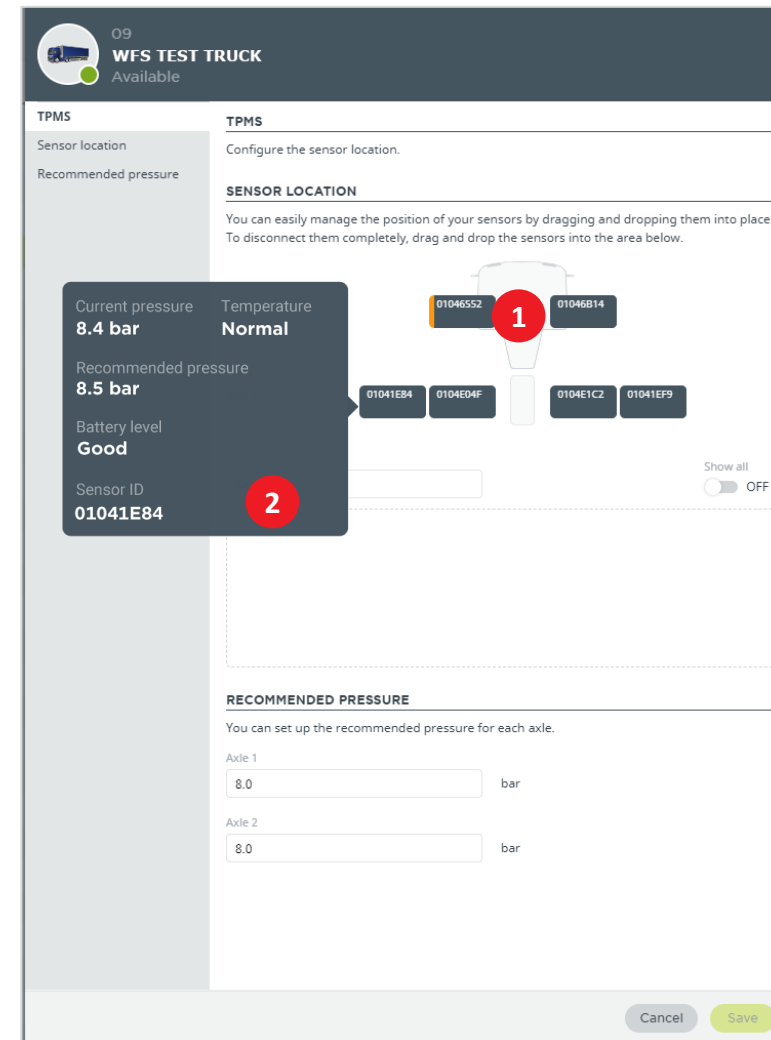
⁽¹⁾ For Manufacturer (OE) TPMS and OEM.connect, the ID shown in WFS does not correspond to the ID physically marked on the sensor. In the case of the Manufacturer (OE) TPMS solution, the ID always starts with the prefix "CAN".



How to edit sensors and set recommended pressure

In Webfleet

- Editing¹ sensors (moving their position on the vehicle chassis; removing or exchanging them) and setting the recommended pressure is generally conducted in the workshop with a technician next to the vehicle.
 - For this purpose, we provide a dedicated Webfleet TPMS Tools App.
- Sensor editing¹ functionality is also available in Webfleet. This covers both workshop use and cases where the technician is unable to use, or does not have access to, the **Webfleet TPMS Tools** app.
 1. The vehicle chassis layout is displayed with a focus on the sensors. The sensor identification number is clearly shown. It's possible to move, remove or exchange sensors using drag and drop. It is also possible to add a new sensor from the fleet sensor pool. New sensors can only be added to the fleet pool using the Webfleet TPMS Tools App.
 2. Hover the mouse over a particular tyre sensor, and a tool tip window displays further information about it.
- The recommended pressure for each axle may be edited in the app where applicable.
 - Manufacturer-fitted sensors and LINK 350 sensors cannot be managed here.



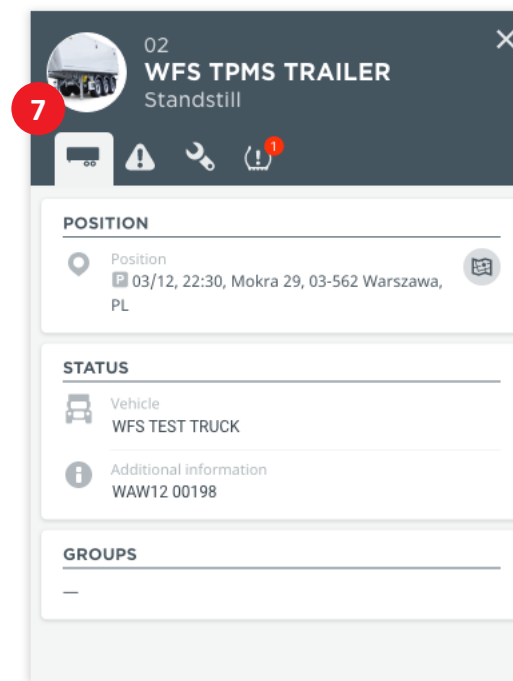
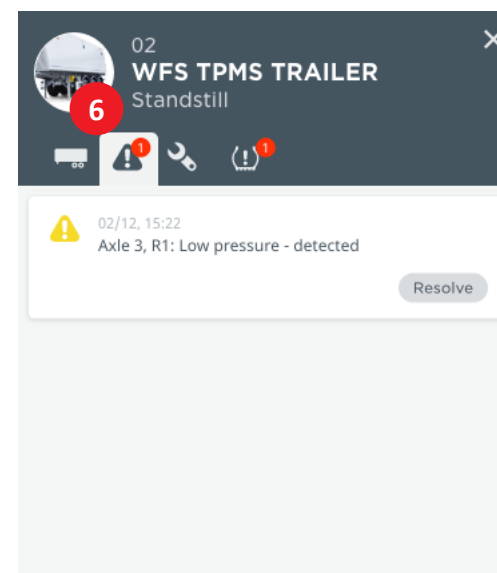
¹ Editing function is not available when sensor data comes from OEM sensors. In that case, the OEM sensors are managed by the OEM/tyre agent.

Detailed vehicle/trailer view

In Webfleet

Under each vehicle or trailer card, alongside the TPMS (!) functionality, you also have:

5. Maintenance tasks
6. Notifications
7. Vehicle/trailer information
 - Last known position*
 - Coupled vehicles/trailers
 - Vehicle/trailer groups



*The last known position of a TPMS trailer is taken from the towing truck. This is only available when the towing truck is also fitted with Webfleet TPMS.

Severity levels display

This is triggered when the value falls below or exceeds a defined threshold*

No issues:
Grey icons



- For the defined threshold details, please see [Appendix](#).
- Where available



Less critical issues: **Amber icons (warnings)**

- Pressure +/- 20% from recommended defaults
- Axle pressure difference: More than 2-bar difference
- Dual wheel pressure difference: More than 1.5-bar difference
- Sensor battery level: Low
- Missing/faulty sensors



Notifications in
Webfleet



Maintenance tasks created in
Webfleet



Critical issues: **Red icons (alarms)**

- Pressure +/- 30% from recommended defaults
- Temperature exceeds 80°C**
- Sensor battery level: Critical**



Notifications in
Webfleet and to
the driver



Maintenance tasks created in
Webfleet

Notifications

In Webfleet/to fleet manager

- Notifications are generated for tyre-related issues according to the tyre status conditions, which includes:
 - Tyre pressure (high/low)
 - Tyre temperature high
 - Dual wheel pressure difference
 - Axle pressure difference

NOTE: When an issue that caused a notification is rectified, a 'back to normal' notification is generated. For example, if a tyre is re-inflated following a low-pressure notification, this message will appear.

The screenshot displays the Webfleet interface. At the top, there's a navigation bar with icons for various functions. Below it, a table lists notifications. The table has columns for Type, Event time, Event description, and Vehicle. The notifications are filtered by 'Warnings' and 'All groups'.

Type	Event ti...	Event	Vehicle
Warning	15/07, 06:37	Axle 2, L1: Low pressure - detected	WFS TEST TRUCK
Notice	13/07, 00:22	Axle 2, L2: Missing sensor - resolved	WFS TEST TRUCK 2
Warning	12/07, 21:49	Axle 3, L1: High pressure - detected	WFS TEST TRUCK
Warning	11/07, 14:02	Axle 2, L2: Missing sensor - detected	WFS TEST TRUCK 6
Alarm 3	11/07, 13:13	Axle 1, L1: Low pressure - detected	WFS TEST TRUCK
Notice	10/07, 16:21	Axle 5, R1: Missing sensor - resolved	WFS TEST TRUCK 2

Below the table, there's a detailed view of a notification for '09 WFS TEST TRUCK'. It shows a warning icon and the text '12/07, 16:22 Axle 2, R1: Low pressure - detected'. A 'Resolve' button is visible. Below this, another notification is shown: '12/07, 11:14 Axle 2, R1: Low pressure - resolved'. This notification is highlighted with a red box, and an arrow points from the text in the note to it. A 'Resolve' button is also present for this notification.

On the right side of the interface, there's a map showing the location of the vehicle. The map is labeled 'Mühlhausen' and shows a road network. A yellow pin indicates the vehicle's location. Below the map, there's a 'WARNING' section with details about the event: 'Event: Axle 2, L1: Low pressure', 'Location: E45/A7, 97230 Estenfeld, DE', 'Event time: 15/07, 06:37', and 'Location time: 15/07, 06:37'. A 'Resolve' button is also present for this warning.

Notifications

To driver

You can choose to provide your drivers with a real-time notification of abnormal tyre issues via the PRO Driver Terminal¹ or in-cab indicator.² This allows them to take action before an issue becomes a serious problem.

How it works: The critical issues that can impact the current journey can either be sent as a simple icon to the driver's PRO Driver Terminal or shown via an in-cab LED indicator. This allows the driver to check the tyres ASAP.

(example)



Critical issues: Red icons (alarms)

- Pressure +/- 30% from recommended defaults
- Temperature exceeds 80°C



TPMS notification in Driver Terminal

TPMS notification in Work App

¹ Available for PRO 2020, PRO 7350, PRO 8375 and PRO 8475

² Availability to be announced

Automated maintenance tasks

In Webfleet

- Maintenance tasks are automatically generated for tyre (sensor) related issues, including:
 - Sensor battery low¹
 - Missing sensor¹
- Missing sensor and low sensor battery maintenance tasks are automatically resolved in cases where the cause of the issue has been cleared.

The screenshot displays the Webfleet MAINTENANCE interface. At the top, there's a navigation bar with the Webfleet logo and 'MAINTENANCE' title. Below it, a filter bar shows 'Due soon tasks' with a '+1' indicator and 'All groups' and 'All vehicles' dropdowns. A table lists maintenance tasks, with one task highlighted: 'Missing sensor' due on '25/06/2020' with a '+49 d' delay. A modal window is open for this task, showing details for vehicle '02 WFS TPMS TRAILER Standstill'. The modal includes a description 'Check vehicle details for more information', due date '16/10/2020' (+52 d), and due (km) as '—'. It also shows 'OVERDUE' status and buttons for 'Details' and 'Mark as read'. The right-hand panel contains sections for 'DETAILS' (Type: General, Task: Missing sensor, Description: Check vehicle details for more information), 'REMINDER' (Date: 18/06/2020, Odometer: —), 'STATUS' (Read: 25/06, 09:33 by Account Administrator, Completed: —), and 'VEHICLE' (Assigned vehicle: WFS TEST TRUCK 2, Position: 19:54, Carrer De Grècia, Constantí (ES 43120), Driver name: Don King, Phone: —).

¹ May not be available for solutions where sensor data is provided by OEM sensors.

Webfleet TPMS Tools app

Why we provide this application

- The background
 - Our TPMS proposition should inform the customer which wheel on the vehicle needs attention.
 - At the point of installation, each sensor will be assigned a position relevant to the vehicle design.
- The challenge
 - Post installation, there are numerous possibilities for the wheel positions to change, e.g.
 - The customer may have their own workshop
 - The vehicle may be maintained by the OEM workshop
 - There may be a contracted/non-contracted tyre service agent
 - There may be a separate emergency tyre service agent
 - There may be a body shop/accident repair centre
 - There may be service scenarios where the wheels are removed from the vehicle, e.g.
 - To service other components such as brakes/suspension/wheel bearings/axel/drive components/bodywork
 - This may involve removing all wheels from the vehicle.
 - It is highly likely that the wheels will not be put back in their original locations. This is particularly the case for trailers.
 - Tyre rotation to account for uneven wear
 - Seasonal tyre changes (summer/winter) where required by legislation
 - Emergency repair

9:41

 webfleet

WEBFLEET account name

Username

Password

In order to start using WEBFLEET, please first agree to our [Terms and Conditions](#).

☐ I have read and agree to these Terms & Conditions.

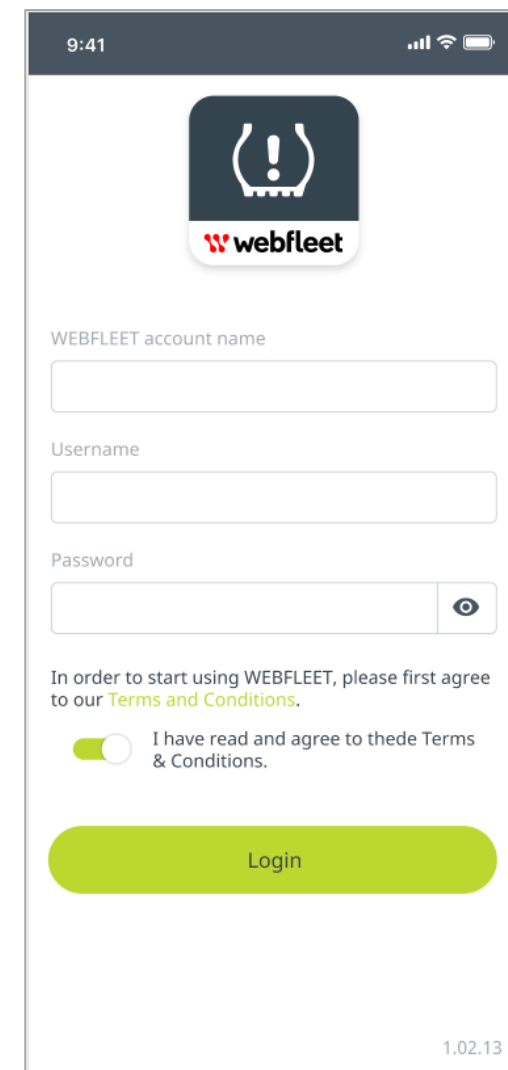
Login

1.02.13

Webfleet TPMS Tools app (continued)

Why we provide this application

- In most of these scenarios, it is unreasonable to expect third parties to either maintain sensor positions or to perform a sensor realignment procedure.
 - Exceptions: The customer has their own workshop or has a close relationship with their tyre service provider.
 - For this reason, we cannot guarantee that wheel positions are correct in Webfleet. It is the customer's responsibility to maintain sensor positions post installation.
- We can, however, provide the customer with tools that make sensor management as simple as possible.¹
 - The Webfleet TPMS Tools app is available for iOS and Android.
 - It is designed to be used in the workshop to easily maintain the correct operation of the Webfleet TPMS system.
 - When wheels are removed from the vehicle (e.g., during service) and subsequently replaced, the TPMS Tools app can be used to ensure that the sensors are shown in the correct position in Webfleet.
 - It can also be used when new sensors are required.
 - A new Webfleet user can be created specifically for users of the TPMS Tools app. This user has restricted access, limited to using this application only.



9:41



WEBFLEET account name

Username

Password

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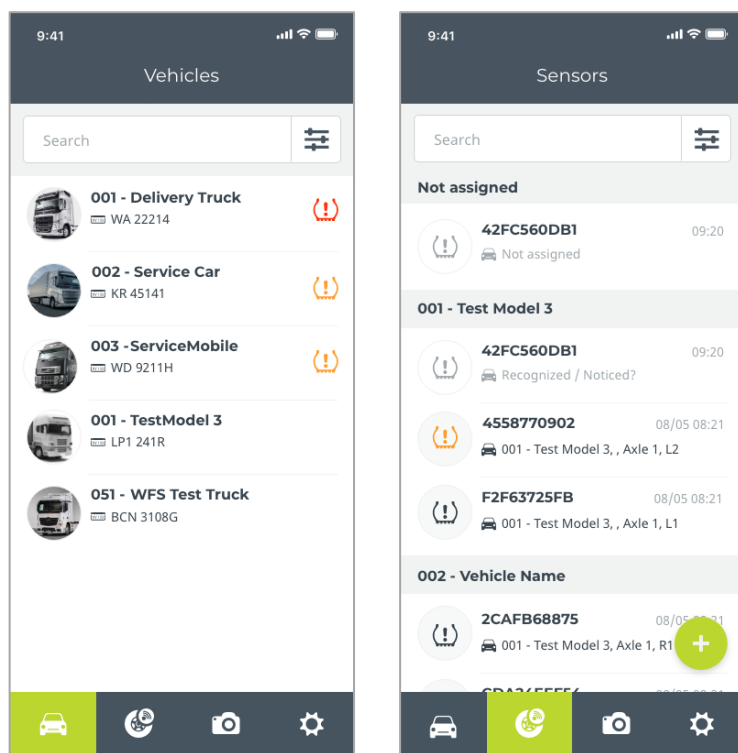
Login

1.02.13

¹ Sensor management is not available when sensor data comes from OEM sensors. In that case, the OEM sensors are managed by the OEM/tyre agent.

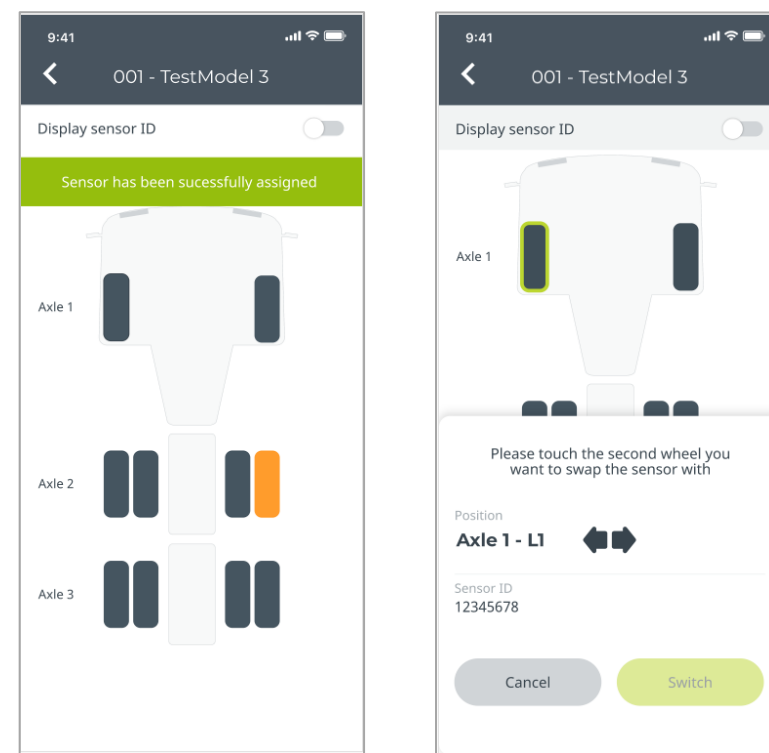
Webfleet TPMS Tools app (continued)

Vehicle centric list available



- All TPMS-equipped vehicles in the fleet are displayed at login.
- You can search for the vehicle that needs to be worked on.

Vehicle view displays the layout of the vehicle chassis



- Any fault conditions are clearly visible.
- Removing, adding and switching actions are executed by touching the tyre positions requiring edits.¹
- Confirmations of changes are also provided to the user.

¹ Editing function is not available for sensor data from OEM sensors or LINK 350. OEM sensors are managed by the OEM/tyre agent and LINK 350 sensors are managed in a specialised application.

TPMS as a modular use

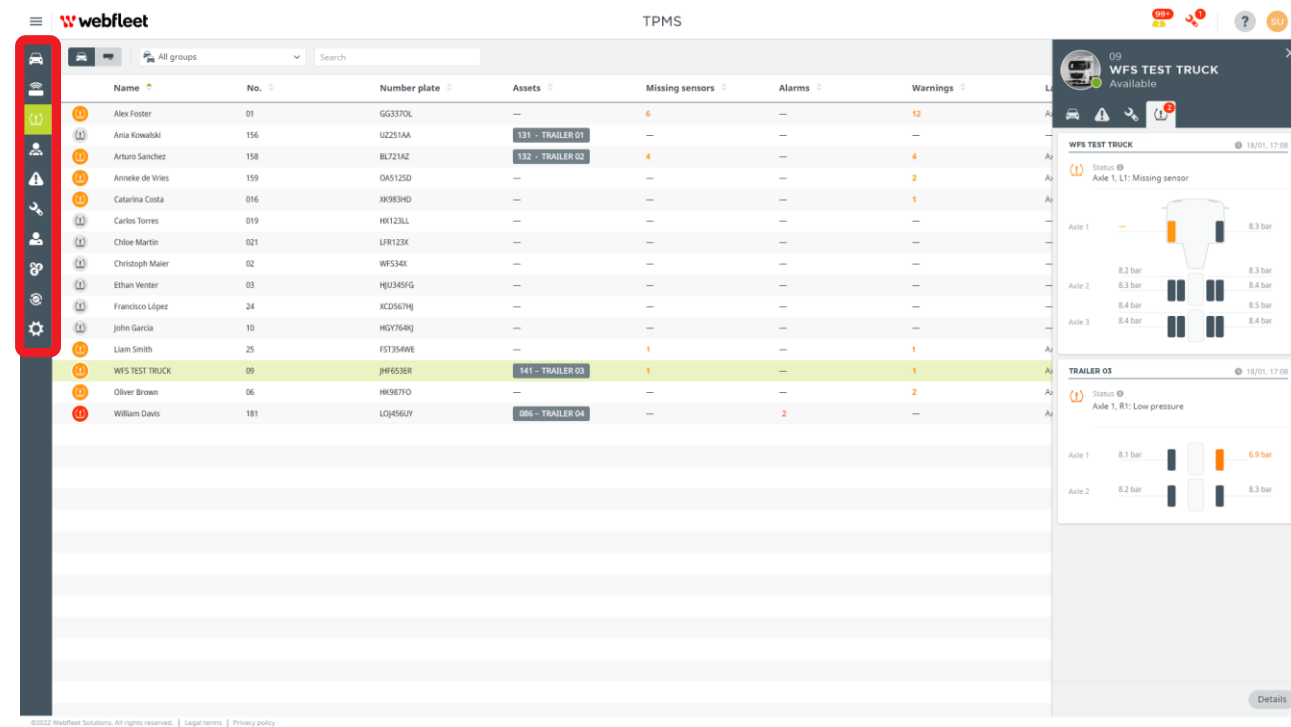
Telematics as you want it: Build your solution to suit your business

- Modular TPMS: Webfleet UI focused on TPMS functionality

- Same TPMS functionality as standard
- No map, trace, trips, OptiDrive, logbook, orders, dashboard, addresses, areas, etc

- Use case:

- Modular option for prospects who have needs of TPMS to improve safety and fleet uptime, but do not want full telematics features or track and trace
- Better serve new customers whose existing telematics providers cannot offer TPMS service
- When ready, the customer can easily upgrade to full Webfleet telematics richness by switching from MOD to ECO or higher tariff and cross-sell to other modular options, e.g., Tacho modular offers.



Modular TPMS UI

Customer benefits



Maximise vehicle uptime

- Reduces breakdown by detecting tyre issues beforehand
- Keeps your service running on schedule with less disruptions due to tyre failure
- Spots issues that are difficult to detect without TPMS, such as slow pressure leaks



Enhance road safety

- Tyres are kept at the correct pressure for optimum handling and braking distance



Lower your running costs

- Properly inflated tyres save fuel. For example, 20% under-inflation causes 2.5% more fuel consumption.
- Correct tyre pressure helps maximise tyre mileage by improving retreadability.
- Maintains your tyres more efficiently with real-time pressure monitoring



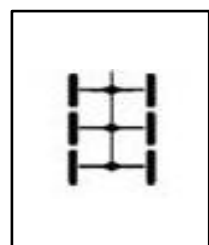
Increase sustainability

- Better fuel economy translates to reduced CO2 emissions.
- Optimises tyre service life and reduces tyre waste

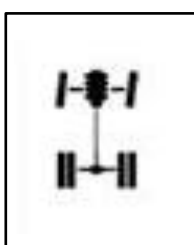


Projected savings

• Typical tractor and trailer configuration



Semi-trailer triaxle



4X2 tractor (truck)

Direct costs

- Tyre purchase price
- Premature removal
- Service life
- Service cost
- Disposal

Indirect costs

- Fuel
- Breakdown (downtime)
- Driver comfort
- Safety

• Background

- Average tyre cost: €400¹
- Excessive wear: Value of the tyre life you'd lose due to underinflation: €200 (cost per remaining projected km)
- 20% underinflation = 2.5% increase in fuel consumption²
- Truck breakdown: Average once every two years¹
 - Ca. €800 each breakdown (incl. new tyre/service call out fees)
 - Average two hours downtime and waterfall costs e.g., penalties and missed opportunities

Calculation (per vehicle per year)

- ❑ The average national haulage truck drives 130,000 kms per year.
- ❑ The average fuel consumption is 30 litres per 100 kms, translating to 39,000 litres per year.
- ❑ The average fuel spend is €78,000 per year (€2 per litre of diesel³).

By eliminating poor tyre inflation with TPMS, a truck can save:

- Fuel (20% underinflation): €78K * 2.5% = **€1950**
- Tyre life: Extended tyre life value = **€200**
- Breakdown/downtime costs:
 - Breakdown fees: **€400** (once every two years)
 - Waterfall downtime costs: **€XXX** (if applicable)

TOTAL SAVINGS:

- **€2550 + €XXX** (if 20% underinflation is prevented)

And, of course, safety is priceless!

¹ Source: Bridgestone

² Internal data from Bridgestone. The impact may be more significant depending on the tyre specifications. 20% under-inflation can increase fuel consumption up to 5% in some older generation tyres.

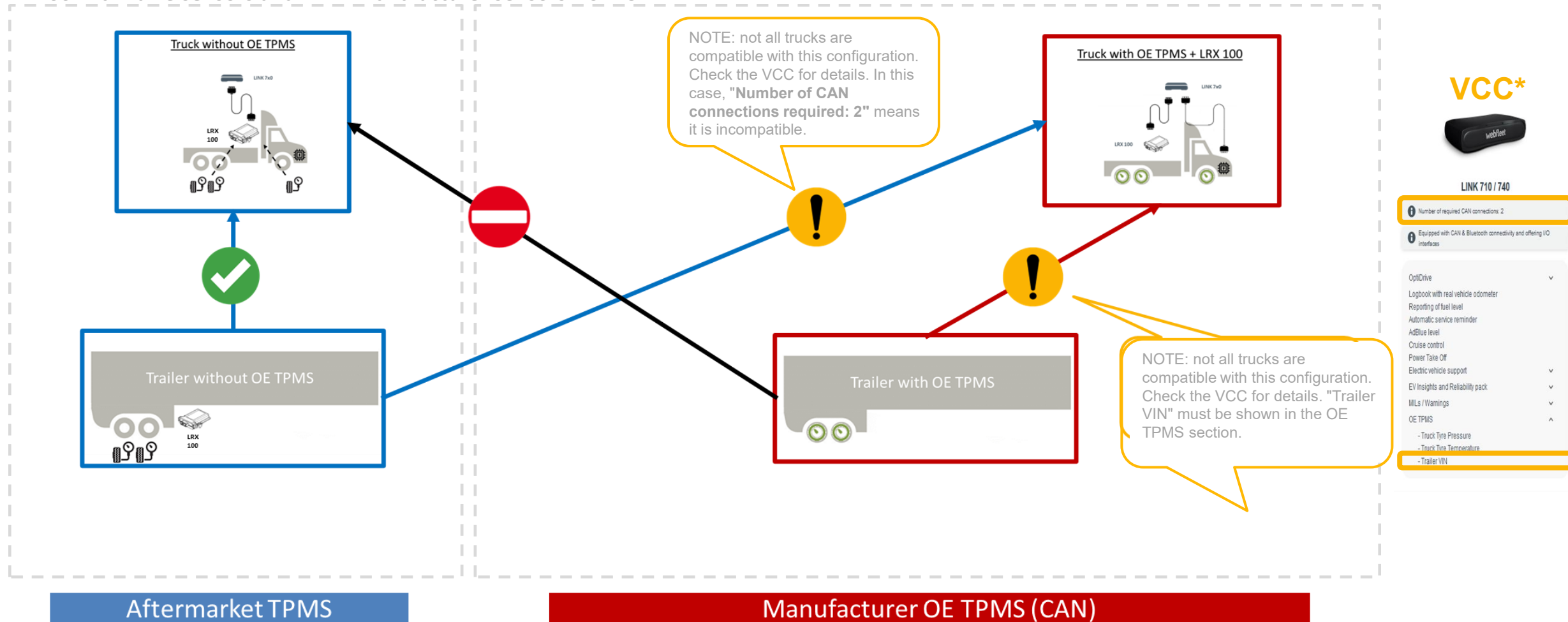
³ Based on EU average Diesel price from 25 Apr 2022.

Product compatibility and requirements

	Hardware Types	Sensor Compatibility	Vehicle Compatibility	Comments
Manufacturer (OE) TPMS	LINK 740/LCS 100 	OEM-fitted sensors (via CAN connection and Direct FMS) 	Bus, truck, truck and trailer 	Refer to Vehicle Compatibility Checker for make/model/year compatibility.
Aftermarket TPMS (valve)	LRX 100 	Valve mount 	Bus, truck, truck and trailer 	For a detailed recommendation of necessary hardware, please refer to the Hardware Calculator .
Aftermarket TPMS (trailer only)	LINK 350 	Strap mount 	Trailer 	For further details, please check the Webfleet Trailer Product Kit .
Manufacturer (OE) TPMS (trailer only)	LINK 350 	OEM-fitted sensors (via EBS port) 	Trailer 	For further details, please check the Webfleet Trailer Product Kit .

Truck/trailer fleet connectivity

LRX 100 with valve sensors and R141 manufacturer sensors from CAN



*Notes for VCC:

- For trucks: VCC indicates the numbers of CAN connections required and which TPMS data is available (refer to "truck tyre pressure/temperature").
- For trailers: the "Trailer VIN" must be available to view TPMS data in Webfleet.

Supported OEM vehicles

Indicative overview. Always refer to **Vehicle Compatibility Check** app ([VCC](#)) to confirm compatibility for specific truck brand/model and related trailer.

 Mercedes-Benz			
MERCEDES-BENZ_ACTROS_(2011-2018) MERCEDES-BENZ_ACTROS_(2018-)_DIESEL MERCEDES-BENZ_AROCS_(2013-2019)_DIESEL MERCEDES-BENZ_AROCS_(2019-) MERCEDES-BENZ_eACTROS_(2021-)_ELECTRIC	DAF_CF_(2018-2022)_ELECTRIC DAF_XB_(2023-)_DIESEL DAF_XD_(2023-)_DIESEL DAF_XF_(2021-) DAF_XG_(2021-) DAF_XG+_(2021-)	SCANIA_SERIES_G_(2017-2024) SCANIA_SERIES_L_(2024-)_DIESEL SCANIA_SERIES_P_(2017-2024) SCANIA_SERIES_P_(2020-2024)_ELECTRIC SCANIA_SERIES_R_(2017-2024)_DIESEL SCANIA_SERIES_R_(2017-2024)_LNG SCANIA_SERIES_R_(2023-2024)_ELECTRIC SCANIA_SERIES_R_(2024-)_DIESEL SCANIA_SERIES_S_(2017-2024) SCANIA_SERIES_S_(2024-)_DIESEL	IVECO_Daily_(2024-)_DIESEL IVECO_Eurocargo_(2024-)_DIESEL IVECO_Stralis_(2011-2021)_CNG IVECO_Stralis_(2013-2018) IVECO_Stralis_(2018-2021)_LNG IVECO_S-Way_(2019-2021) IVECO_S-Way_(2019-2024)_LNG IVECO_S-Way_(2021-2024)_DIESEL IVECO_S-Way_(2024-)_DIESEL
			
MAN_TGM_(2024-)_DIESEL MAN_TGS_(2024-)_DIESEL MAN_TGX_(2013-2020) MAN_TGX_(2024-)_DIESEL	VOLVO_FH_(2013-2020)_DIESEL VOLVO_FH_(2020-) VOLVO_FH_(2021-)_ELECTRIC VOLVO_FH_(2023-)_LNG VOLVO_FL_(2024-)_DIESEL VOLVO_FM_(2021-)_ELECTRIC	RENAULT_Series_T_(2013-2021) RENAULT_Series_T_(2024)_DIESEL	FORD_F-Line_(2024-)_DIESEL FORD_F-MAX_(2023-)_DIESEL

Commercial terms

Truck/bus



Trailer



Trucks **WITHOUT** manufacturer TPMS

Hardware	Purchase price
LINK 740	€319.00/£279.00
LRX 100 TPMS Receiver	€169.00/£149.00
TPMS 4x2 Vehicle Sensors (6)	€219.00/£189.00

Tariff	Price
ECO or higher	€18.00/£13.50 month
Feature	Price
Webfleet TPMS	€3.95/£3.35 month

Trucks **WITH** manufacturer TPMS (R141)

Hardware	Purchase price
LINK 740	€319.00/£279.00
LCS 100 w/o power cable	€25.00/£21.00
LRX 100 TPMS Receiver*	€169.00/£149.00

*optional LRX 100, required only for connectivity with LRX 100 trailers

Tariff	Price
ECO or higher	€18.00/£13.50 month
Feature	Price
Webfleet TPMS	€3.95/£3.35 month
WF Direct FMS	€3.45/£2.50 month



Trailers **WITHOUT** manufacturer TPMS

Hardware	Purchase price
LRX 100 TPMS Receiver	€169.00/£149.00
TPMS Rigid Sensors (2) (TPMS Rigid Sensors (2) x 3)	€69.00/£59.00 each (€207.00/£177.00 total)
Tariff	Price
ADM	€0.00/£0.00 month
Feature	Price
Webfleet TPMS	€3.95/£3.35 month

Trailers **WITH** manufacturer TPMS (R141)

Incompatible

Trailers **WITHOUT** manufacturer TPMS

Hardware	Purchase price
LRX 100 TPMS Receiver	€169.00/£149.00
TPMS Rigid Sensors (2) (TPMS Rigid Sensors (2) x 3)	€69.00/£59.00 each (€207.00/£177.00 total)
Tariff	Price
ADM	€0.00/£0.00 month
Feature	Price
Webfleet TPMS	€3.95/£3.35 month

Trailers **WITH** manufacturer TPMS (R141)

Hardware	Purchase price
No hardware device required	
Tariff	Price
ADM	€0.00/£0.00 month
Feature	Price
Webfleet TPMS	€3.95/£3.35 month

Trailer TPMS with LINK 350

Now available for trailers WITH manufacturer TPMS sensors. No need to supply/fit sensors.

Trailer TPMS with OEM.Connect

Requires active subscription to OEM service.



Hardware	Purchase price
LINK 350 and Bracket (x1)	€311.00/£269.50
LINK 350 SMS TPMS Internal Sensor (6x)	€240.00/£210.00
LINK 350 SMS TPMS Fastening Strap 22.5"	€60.00/£51.00
Tariff	Price
AST-T	€9.95/£7.95 month
Feature	Price
Webfleet TPMS	€3.95/£3.35 month

Hardware	Purchase price
No hardware device required	
Tariff	Price
AST-T	€9.95/£7.95 month
Feature	Price
Webfleet TPMS	€3.95/£3.35 month

Note: please refer to the latest version of the price list for the most up-to-date pricing information. Prices are subject to change.

For further details, please check the Webfleet Trailer Product Kit.

Key role responsibilities for manufacturer TPMS installation

To ensure an efficient process from sale to setup, we need clearly defined responsibilities for all involved parties.

CUSTOMER

- Must detail which vehicles fitted with manufacturer TPMS at point of sale.
- Must take responsibility for the correct data entry/onboarding of vehicles via Webfleet in the case of trailers fitted with manufacturer TPMS but no LINK device.
- Must maintain a subscription to the OEM service provider for trailers that will use OEM.connect.
- Must detail the EBS system per trailer and port activation status at point of sale in the case of trailers that will have LINK 350 fitted.
- Must take responsibility for maintaining the TPMS system as part of general vehicle maintenance (including tyre rotations/replacements/repairs...) for all vehicles with manufacturer TPMS.
- Must inform the sales team if any vehicles in the fleet must comply with the ADR agreement/regulations.
- Must ensure the party responsible for tyre maintenance is properly instructed/informed about installed systems.
- Must ensure tyres are correctly inflated after the installation of TPMS systems.

SALES TEAM

- Must ensure the customer understands their responsibilities.
- Must complete the installation checklist Excel accurately, detailing the type of TPMS solution and components required (including service/spare components).
- Must check fleet compatibility, with particular attention to truck/trailer solutions—not all new trucks with manufacturer TPMS can support older trailers with the LRX aftermarket solution. In these fleets, either compatibility with only new trailers is accepted or a LINK 350 is required for trailers.
- Must engage a TPMS-certified sales engineer/field auditor/field engineer for all TPMS solutions to support during the pre-sales process.
- Must provide warning label in relevant languages for aftermarket valve sensor solution.
- Must ensure customer is trained to understand and act upon TPMS insights.

INSTALLATION TEAM

- Must check whether vehicle has TPMS installed by the manufacturer before any installation is started.
- Must check the correct solution will be installed—aftermarket TPMS sensors are not to be installed where TPMS sensors from the manufacturer are already fitted.
- Must report any discrepancies/changes/errors, such as incorrect solution booked or a defective manufacturer TPMS system.
- Must ensure they are informed of the latest installation procedures for TPMS solutions.

Essential components for aftermarket TPMS (valve-mounted sensor)

Calculate the hardware components and quantities you require by entering a few simple details in our Hardware Calculator .

Benefits:

- Covers 90% of use cases
- Easy to use and validate
- Helps to order accurately
- Connected to SRP price list to generate order form

How to use the Hardware Calculator:

Step ① Input: # Vehicles

Step ② Input: # Surplus ratio → total components*

Step ③ Order configuration of Webfleet packages

Step ④ Order form

COUNTRY

CURRENCY

VAT

Australia

Australia

Belgium

CzechRepublic

Denmark

France

Germany

Ireland

Italy



Webfleet Solutions Sales B.V.
Beethovenstraat 503, 1083 HK Amsterdam,
The Netherlands
VAT No BE 0552.719.757

PRODUCT DESCRIPTION	ARTICLE NR	Qty	SRP incl VAT	SRP ex VAT	TOTAL SRP ex VAT
LRX 100	1KTM.002.00.1	40	-	-	-
TPMS 4x2 Vehicle Sensors (6)	9KTS.002.41.1	0	-	-	-
TPMS Twin Wheel Axle Sensors (4)	9KTS.002.41.3	1	-	-	-
TPMS Rigid Sensors + Angled Extn (2)	9KTS.002.15.1	11	-	-	-
TPMS Rigid Sensors (2)	9KTS.002.10.1	11	-	-	-
TPMS Rigid Sensors (56)	9KTS.002.16.1	1	-	-	-
TPMS Rigid Sensors (112)	9KTS.002.15.3	2	-	-	-
TPMS Short Flex Sensors (16)	9KTS.002.21.1	0	-	-	-
TPMS Long Flex Sensors (16)	9KTS.002.31.1	0	-	-	-
TPMS Installation Tool Set	9KTS.002.65.1	1	-	-	-
LRX 100 Cable	9KTM.002.00.2	0	-	-	-
TPMS Sensor Seals (100)	9KTS.002.70.1	1	-	-	-
TPMS Plastic Sensor Extn (50)	9KTS.002.77.1	2	-	-	-
TPMS Angled Extn (50)	9KTS.002.11.2	2	-	-	-
TPMS Sensor Clamp Alloy Wheel (50)	9KTS.002.61.1	0	-	-	-
TPMS Sensor Clamp Steel Wheel (50)	9KTS.002.51.1	0	-	-	-
LINK 7x0 TPMS Warning LED	9KX4.001.30.1	0	-	-	-
TPMS OTR Sensor Pack (2)	9KTS.002.12.1	0	-	-	-
TPMS OTR Sensor Pack (84)	9KTS.002.12.2	0	-	-	-
OTR Sensor Seals TPMS (x100)	9KTS.002.73.1	0	-	-	-
Total Hardware Price		72			0
					-

Hardware calculator step 4

*TPMS sensor packages cover most of the use cases, but special wheel rim configurations/design may require a different type of angled extension and/or additional components. For these cases, we highly recommend involving a Webfleet sales engineer or Bridgestone field engineer teams to verify your configuration.

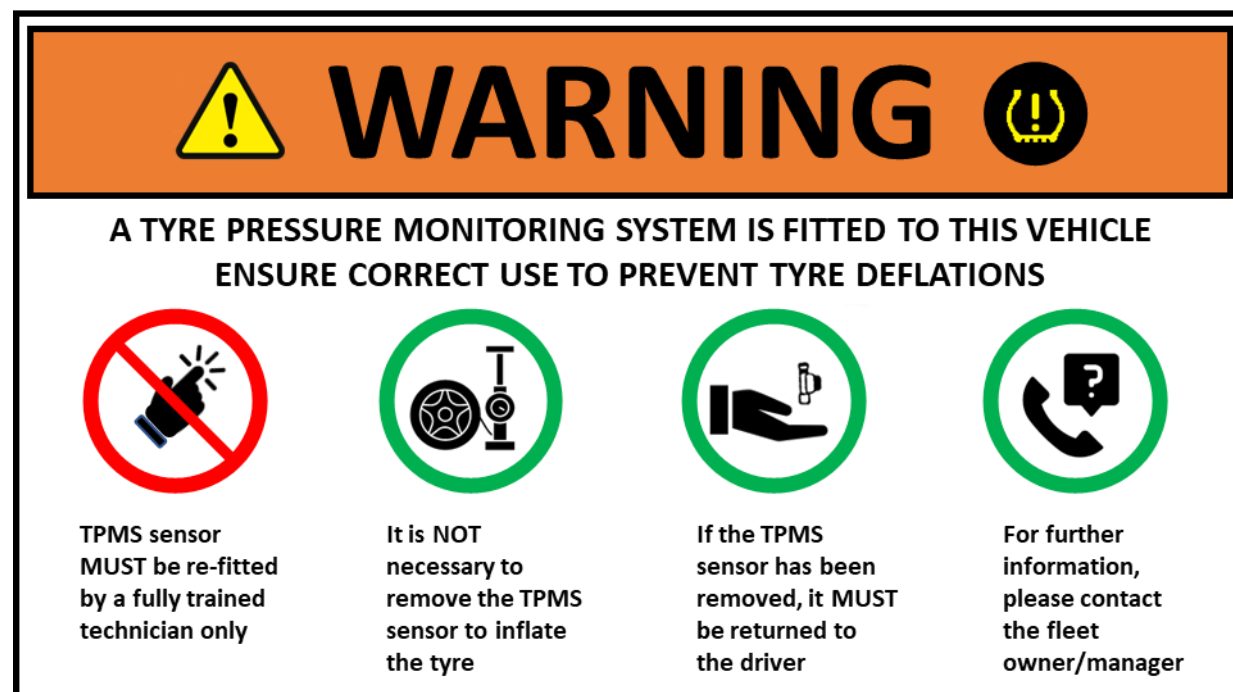
Trailers with Manufacturer (OE) TPMS

In Webfleet

- New trailers fitted with TPMS by the OEM and complying with UN ECE R141 do not require any additional hardware to be fitted
- These trailers will send TPMS data to the towing tractor once physically connected as part of UN ECE R141
- Where the towing truck is also UN ECE R141 compliant, equipped with Webfleet TPMS and Direct FMS, the trailer data will be sent to the Webfleet cloud via the truck (LINK 740)
- Such trailers are onboarded via the Webfleet user interface

Warning label

- Why do we have a warning label?
 - The valve stem TPMS sensors must be installed and maintained by trained professional, using the correct tools and process.
 - Maintenance/interference with the sensor by untrained operatives has a high probability of causing pressure loss.
 - For this reason, it is important that the person responsible for the day-to-day care of the vehicle understands how to manage sensors if they need to be removed.
- Where can the warning label be found?
 - <https://portals.webfleet.com/s/article/TPMS-Sensor-Warning?>
 - It is available in multiple languages.



Trailers with Schmitz Cargobull telematics

OEM.connect TPMS (VIA Schmitz Cargobull)



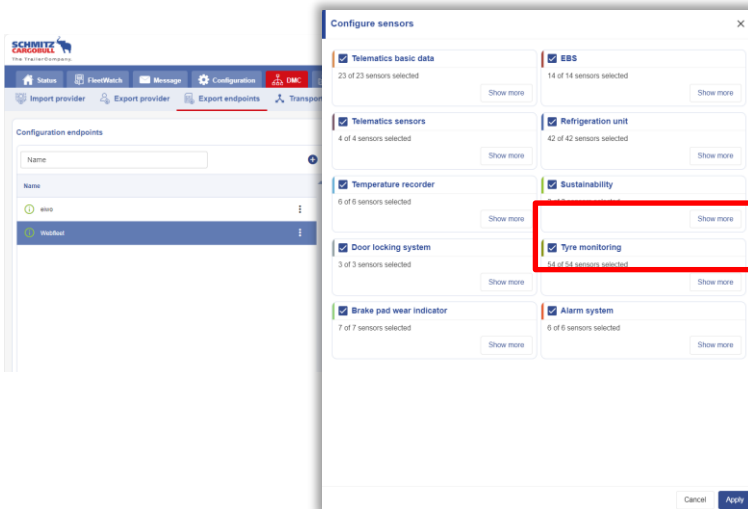
A full description can be found in the Webfleet Trailer Product Kit.

- OEM-connected asset data coming via Schmitz Cargobull connected trailers.
- This integration connects customers with EXISTING access to “TrailerConnect” who choose Webfleet as an endpoint for their fleet data.
- Depending on the factory-fitted telematics from the OEM, customers can choose several datasets to send to Webfleet– **including TPMS when available.**



Webfleet Trailer’s integration of Schmitz Cargobull telematics connects with the existing feature sets:

1. Basic telematics data
2. TPMS
3. Cold Chain (1-wire sensor visualisation)



Technical specifications

1 LRX 100 – TPMS Receiver



2 TPMS Sensor



LRX 100	
Dimensions	135 x 120 x 35 mm
Weight	220g
Protection class	IP 69K
Supply voltage	12 V / 24 V (min. 8 V to max. 32 V)
Power consumption	700mW Max.
Temperature	Operation: -40 °C to +85 °C Storage: -40 °C to +85 °C
Sensor – LRX 100 operating frequency band	433.05 - 434.79 MHz
LRX 100 Primary – Auxiliary operating frequency band	2400-2483.5 GHz
Sensor	
Dimensions	40 x 23 x 20.5 mm
Weight	30g
Temperature	Operation: -30 °C to +85 °C Storage: -30 °C to +85 °C
Pressure range	0-12 bar
Battery life	3 years
Sensor – LRX 100 operating frequency band	433.05 - 434.79 MHz
Accuracy	Typical: +/- 0.2 bar; max: +/- 0.5 bar

The details listed in the product specification are intended to provide a general description of the product and are subject to change.

Technical specifications: LINK350



Compliance

- CE compliant
- ISO 16750
- ADR (EX/II, AT)
- LINK 350 currently getting certified (ATEX)

Communicating

- Internal GPS / Glonass antenna
- Integrated GPS and GSM module
- Embedded SIM (EU28)
- 433 MHz receiver for wireless TPMS sensors

Connecting

- Wabco, Haldex, Knorr EBS
- Direct tyre pressure via embedded ECU

Operating

- Two-point fixation (under trailer, between axles)
- IP 6K9K: fully dust-tight and high-pressure waterproof
- -40°C to + 75°C

Charging

- Single connectivity to T-EBS
- Rechargeable lithium-ion 3.6V
- 180 events sent per charge, over 30 days at room temperature
- Charging mode requires temperature to be with 0°C to +/- 45°C

Tyre status conditions

- Notifications are generated under various conditions, the following table describes the default behaviour

Issue	Warning threshold Amber 	Alarm threshold Red (1 st alarm) 	Alarm threshold Red (2 nd alarm) 	Webfleet notification ¹   	Driver notification 	Remarks
Low pressure	20% below recommended pressure	30% below recommended pressure	40% below recommended pressure	✓	✓ Alarms only	Configurable
High pressure	20% above recommended pressure	30% above recommended pressure	40% above recommended pressure	✓	✓ Alarms only	Configurable
High temperature	N/A	Above 80° C	N/A	✓	✓ Alarms only	Configurable ²
Axle pressure difference	More than 2-bar difference	N/A	N/A	✓	✗	Configurable ²
Dual wheel pressure difference	More than 1.5-bar difference	N/A	N/A	✓	✗	Configurable ²

- Maintenance tasks are generated for low sensor battery and missing sensors

Issue	Amber	Red	Maintenance task ³	Remarks
Sensor battery**	Level: Low	Level: Critical	✓	Current sensor provides a status of good, low or critical. Battery low (amber) with due date of seven days; battery critical (red) with due date of one day.
Missing/faulty sensor**	Sensor is not seen for a period of one hour (during which time other sensors are seen)	N/A	✓	

¹ When an issue that caused a notification is rectified, a 'back to normal' notification will be generated (Notice). For example, this will occur if a tyre is re-inflated following a low-pressure notification.

² Can be configured at launch only via Webfleet Customer Support

³ Missing sensor and low sensor battery maintenance tasks are automatically resolved in cases where the cause is cleared.

**Where available

Sales order process: identifying the right solution

- Understanding the specific requirements of each vehicle is essential for successful implementation.
- Determining whether a vehicle has TPMS fitted by the OEM is the first important step.
- The customer must provide you with accurate vehicle information, but you can confirm it yourself based on the information below:
 - Date of first registration: all commercial vehicles registered after 7 July 2024 are required by EU regulation R141 to have TPMS fitted by the manufacturer.
 - Date of manufacture: the year of manufacture may be determined by the VIN. The year is represented by character 10. "S" represents 2025, 'T' represents 2026, etc.
 - Customer's invoice/contract/order for the vehicle: vehicle specifications are usually stated on their sales documentation.
 - OEM dealer relationship: OEM dealers for each vehicle brand can use internal systems to check vehicle specifications based on VIN.
 - Physical inspection: using specially designed tools (ATEQ/Bartec), it is possible to determine if a vehicle is fitted with a TPMS sensor; these tools are typically used by tyre agents and Market Technical Team engineers.
- Installation order form
 - The Installation Checklist Excel has been updated to reflect the new configurations.
- Sales engineering
 - We recommend you involve your sales engineer for TPMS solutions.

webfleet solutions
a Bridgestone company

Webfleet Contract No

Field Service Associate / Project Manager (if applicable)

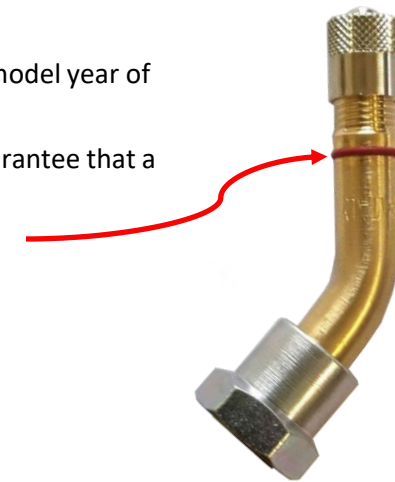
Installation Order Form

Customer Information

Contact Person Name Contact Person Company Contact Person Phone Contact Person Mobile Contact Person Email

High-level installation process: Manufacturer TPMS

- Before attempting to install *any* TPMS solution, the vehicle must be checked for OE-fitted TPMS. Since July 2024, all new EU vehicles in categories M1, M2, M3, N1, N2, N3, O3 and O4 (~truck/bus/trailer) must be equipped with a functioning Tyre Pressure Monitoring System. However, many vehicles sold before this date are compatible.
 - To identify a compatible vehicle, first use the Vehicle Compatibility Checker. (The right CAN configuration is essential for correct operation.)
 - This will identify that the vehicle generation (make/model/year) is compatible but cannot determine if TPMS is fitted.
 - To determine if TPMS is fitted:
 - Check the VIN plate for year of manufacture—if character 10 is "S" or greater, Manufacturer TPMS can be fitted. All European commercial vehicles with a model year of 2025 and above will have TPMS fitted. (Check the dashboard for warning lamps and/or TPMS settings on the vehicle configuration screens.)
 - Tyre and Rim Technical Organisation (ETRTO) recommends that valves are marked where a TPMS sensor is present. This is not mandatory and does not guarantee that a sensor is fitted, but it is a good guide. Valves are usually marked with a coloured O-ring or similar.
 - Check for OE sensors with a dedicated device, such as those available from ATEQ or Bartec.
- Prerequisites for a complete installation:
 - LINK 740/710 with minimum firmware version 13.2. For trailer support, 13.6 or greater is required.
 - Webfleet contract with the TPMS feature and Direct FMS feature enabled and assigned.
 - Correct CAN configuration assigned to the vehicle.
 - Vehicle set up using Installer App (version 2.1 or higher).
 - Installer App user with the relevant user rights.
 - DIP switches correctly positioned on the LINK device.



Note: Tyres must be inflated to the correct operating pressure for the vehicle before starting any TPMS installation.

High-level installation process: Webfleet Hardware

- To install TPMS sensors correctly, all valves, wheel rims and tyres must be in good condition. Customers should be advised that:
 - Wheel rims and valves must be clean and in serviceable condition. The valves must be the correct type for the wheel rim so the sensor does not touch other parts or extend beyond the tyre width. Tyres must be inflated to the correct operating pressure for the vehicle.
- Once trailers are fitted with TPMS, a Webfleet TPMS tractor unit should be made available to test and validate the installation.
- We recommend the following sequence of steps for an optimal installation procedure:
 - Enable the LINK device for TPMS functionality. Check for the correct firmware. Adjust the DIP switches on the LINK device.
 - Provisionally mount the LRX 100 and run the cable to the LINK device.
 - Install the auxiliary LRX 100, if required.
 - Set up the vehicle in the LINK Toolkit App.
 - Mount the sensors and register them with the LINK Toolkit App.
 - Check the signal strength in the LINK Toolkit App and adjust the position of the LRX 100 for optimum signal strength.
 - Fix cables, finalise installation and test functioning.
- Prerequisites for a complete installation:
 - LINK 740/710 with firmware version 8.13 or higher.
 - Webfleet contract with the TPMS feature enabled.
 - LINK Toolkit App (version 2.4 or higher).
 - LINK Toolkit user with the relevant user rights.

Note: test the installation of the TPMS receiver(s) and sensors. Both must function outside of the garage with a minimum distance of 20m from surrounding buildings and walls.

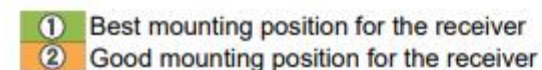
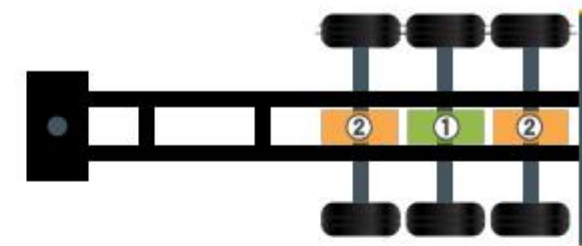
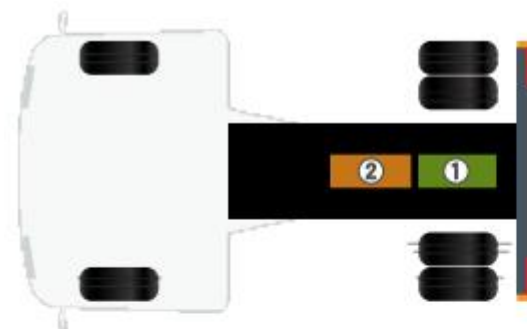
Installation spotlight items

- Customer prerequisites

- Wheel rims and valves must be clean and in serviceable condition. The valves must be of the correct type for the wheel rim, so the sensor does not contact other parts or extend beyond the tyre width.
- Tyres must be inflated to the correct operating pressure for the vehicle.

- Receiver box

- The correct placement of the LRX 100 is essential to receive the best signal. Try to avoid metal shielding, if possible, to have free sight between the receiver and sensors.
- Check the power supply and make sure the primary/auxiliary LRX 100 connection is correct.



Installation spotlight items

- Sensors

- 1 The most important element is that the sensor doesn't stick out, as it could easily be torn off and broken.
- 2 The sensor's position shouldn't block tyre inflation, as inflation is still possible after the sensors are installed.
- 3 Incorrect sensor installation may cause tyre air leakage. Also, you will need to restart installation.
- 4 If there is not enough space for sensors, the sensors won't work and the risk of breaking the valve sensor or the valve tyre is high.



Vehicle support examples

- Compatible: Solutions available for standard HCV (heavy commercial vehicles)



Rigid-body truck



HCV tractor



Trailer



HCV tractor trailer combination



City bus, touring coach or articulated city bus

- Incompatible or requiring validation before order/fitment



Light commercial vehicles



Passenger cars



Hazardous goods/ADR trailer*



Any vehicle with CTIS/TPRS



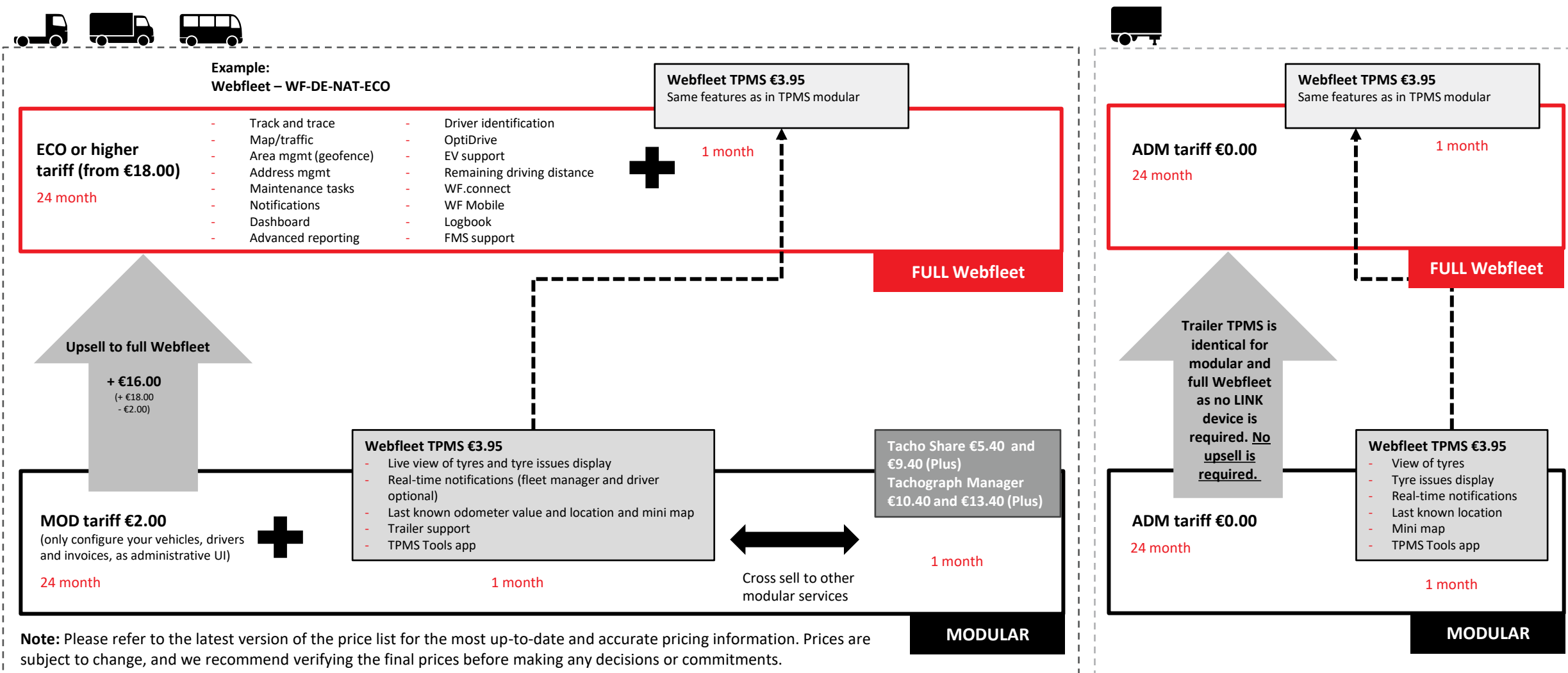
Plant/machinery



Custom-built HCV

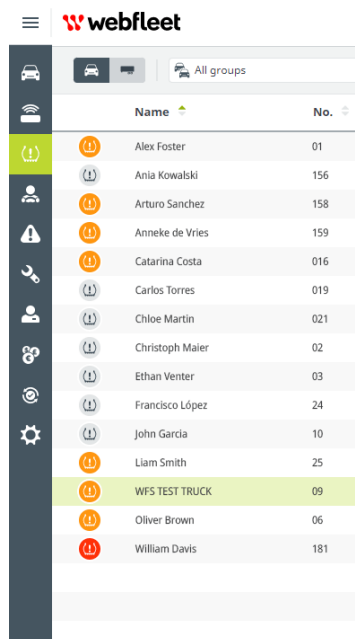
*The LINK 350, which has been formally certified by the supplier for **some levels of ADR compliance**. Supported vehicles³: ADR EX/II (explosive materials) or EX/III (very explosive materials) or AT (this refers to a vehicle, other than EX/III, FL or OX vehicle).

Upgrade paths: Modular cross-sell and upgrade to full Webfleet



User interface comparison

- TPMS service modular versus full Webfleet telematics (starting at ECO tariff)



- ✓ Included
- ✓* Limited
- ✓ Module remains, but no data
- ✗ Not included
- + Optional

* MOD tariff plus TPMS additional service:

- Some modules have limited functionality:
 - Vehicles (no trace and trips)
 - Drivers (no OptiDrive)
 - Reports (only relevant)

¹ Last position available under Vehicle view

Feature Set	MOD + TPMS Modular	ECO + TPMS full Webfleet
Compatible with full Webfleet*	✗	✓
Map	✗ ¹	✓
Vehicle	✓*	✓
Asset tracking	+	+
TPMS	✓	✓
Driver	✓*	✓
OptiDrive	✗	✓
Order	✗	✓
Notifications	✓	✓
Video	✗	+
Checklists	✗	+
Maintenance	✓	✓
Dashboard	✗	✓
Reports	✓*	✓
Tachograph	+	+
Logbook	✗	✓
Addresses	✗	✓
Areas	✗	✓
Users	✓	✓
Invoices	✓	✓
Administration	✓	✓
Settings	✓	✓
Webfleet Mobile	✗	✓
Logbook App	✗	✓
Work App	✗	✗
Vehicle Check app	✗	+
TPMS Tools app	✓	✓

